



MASTER 1 THESIS — GEOMATICS AND SPATIAL MODELLING

Mapping Urban Heat Islands in Aix-en-Provence

Remote sensing and Local Climate Zones (LCZ) — heatwave of 16 August 2025

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01

Context and Problem Statement

Urbanization, surface energy balance alteration, and microclimatic vulnerability in Aix-en-Provence.

CONTEXT

Study area: the commune of Aix-en-Provence

Growing urbanization replaces natural soils with mineral surfaces, reducing vegetation and altering the surface energy balance. This drives the urban heat island (UHI) effect, with higher temperatures in cities than in the rural periphery (Oke, 1982).

The commune of Aix-en-Provence (186 km²), under a Mediterranean climate marked by hot, dry summers, combines a dense historic centre, an expanding residential periphery, and extensive natural and agricultural areas (garrigue, olive groves, wooded massifs).

186 km²

Municipal area analysed, from the dense historic centre to garrigue

40.1 °C

Maximum air temperature recorded on 16 August 2025 (Météo-France)

210,036

30 m cells forming the municipal analysis grid

CONTEXT

Location within the regional context

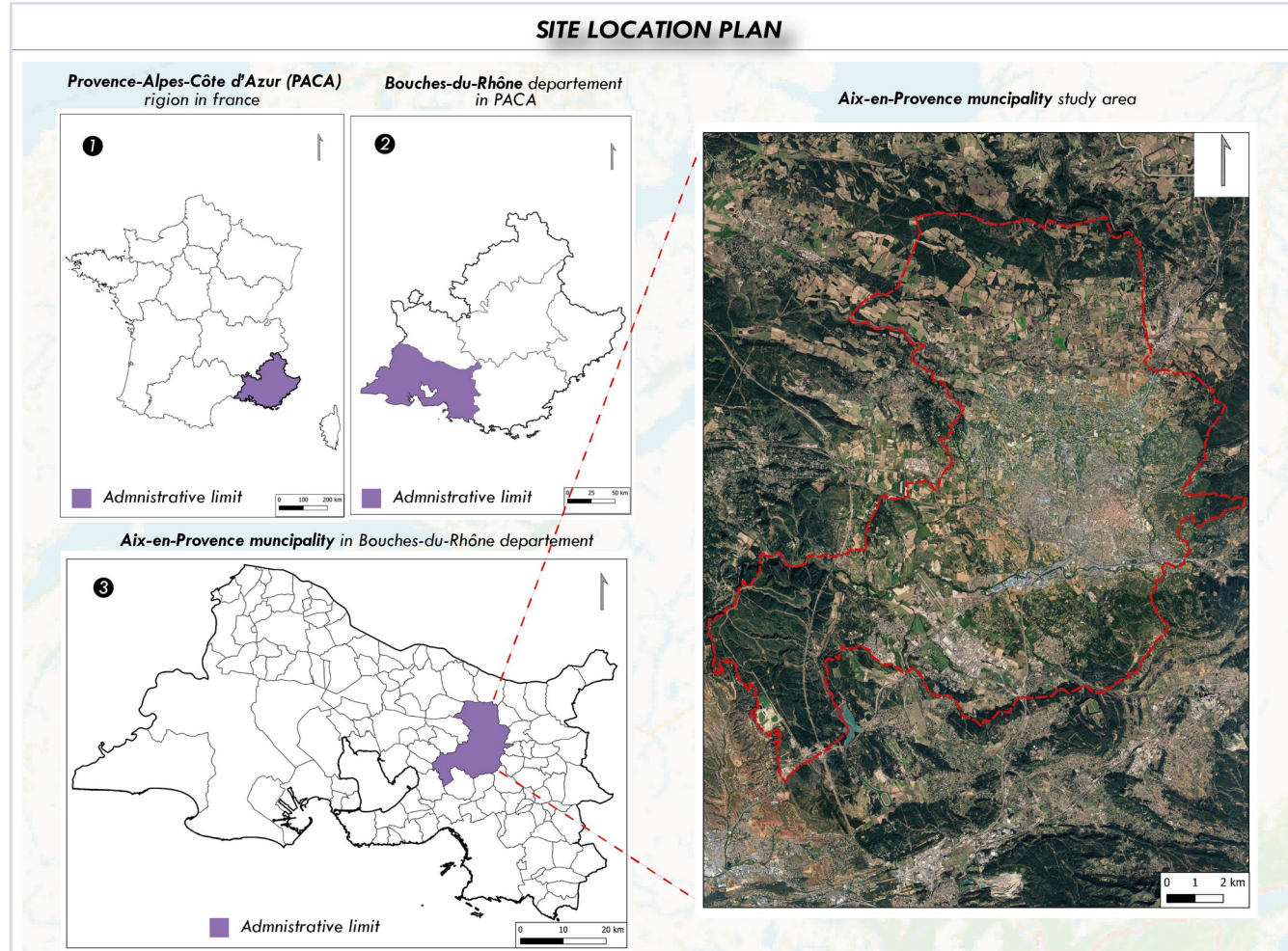


Figure 1 — Location of Aix-en-Provence in its regional context (Bouches-du-Rhône / PACA)

The commune of Aix-en-Provence (INSEE code 13001) sits within the Bouches-du-Rhône department, in the Provence-Alpes-Côte d'Azur region.

The satellite view on the right shows the administrative boundary used to clip the analysis grid: a dense urban core surrounded by agricultural plains to the north and wooded, hilly terrain to the south and west.

This morphological diversity — from a historic centre to open garrigue — makes the commune a relevant laboratory for studying intra-urban thermal contrasts.

Source: IGN administrative boundaries; satellite basemap.

The Local Climate Zone (LCZ) system

Proposed by Stewart & Oke (2012), the LCZ system defines 17 standardized climate classes based on measurable morphological properties — building density, height, vegetation cover, roughness — independent of urban function (residential, industrial, commercial).

Built classes — LCZ 1 to 10

From very dense, tall fabrics (LCZ 1 — Compact high-rise) to very open, low ground-coverage fabrics (LCZ 9 — Sparsely built).

Each class is defined by a characteristic combination of mean building height, building surface fraction, and impervious surface fraction.

Natural classes — LCZ A to G

Seven classes covering wooded surfaces, low plants, bare soil, rock, and water.

It is this logic of classification by morphological thresholds, rather than by land use, that was reproduced in this study.

Objective

1. Map land surface temperature (LST) in Aix-en-Provence during a heatwave, using Landsat 8.
2. Classify the city into Local Climate Zones (LCZ) at 30m resolution.
3. Quantify, via regression, how building density, vegetation, height, and roughness explain surface temperature.

02

Data and Methodology

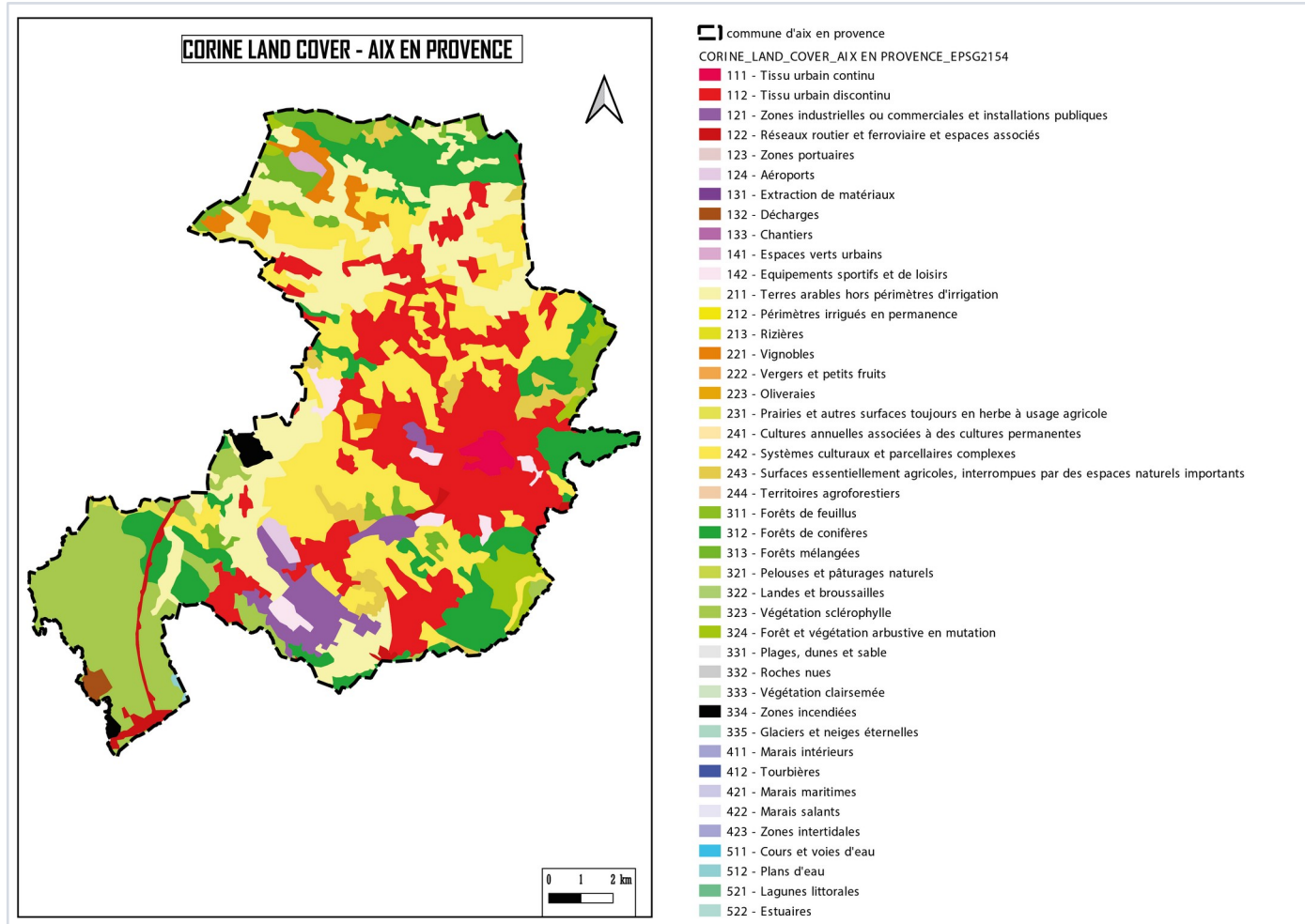
From satellite imagery preprocessing to multiple linear regression modelling.

Geospatial datasets used

All processing was carried out in Google Colab, coupled with Google Earth Engine for raster data and the geopandas, rasterstats, and statsmodels libraries for vector analysis and statistical modelling.

METHODOLOGY

Corine Land Cover: correcting BD TOPO gaps



Corine Land Cover — Aix-en-Provence (Copernicus, 100 m)

Since BD TOPO coverage presents localized gaps, cells initially classified as "F — bare soil" due to missing data rather than actual ground conditions were corrected using the dominant Corine Land Cover class.

The map shows a dense continuous/discontinuous urban fabric (111/112, in red) at the centre, surrounded by a mosaic of arable land and complex cultivation patterns (211/242, in yellow), with broadleaved and coniferous forests (311/312, in green) dominating the northern and south-western parts of the commune.

Source: Copernicus Corine Land Cover, reclassified via a CLC → LCZ correspondence table.

Analysis grid and morphological indicators

A regular 30 m grid — aligned with the native resolution of the Landsat 8 thermal sensor — was built over the whole commune (210,036 cells). For each cell, five indicators are computed by geometric intersection with the reference vector layers.

Building fraction

Proportion of a cell's surface occupied by building footprints — the main descriptor of ground density.

Vegetation fraction

Proportion of vegetation cover — a cooling factor through evapotranspiration and shading.

Impervious fraction

Sum of building and road surfaces, capped at 1 to correct geometric overlap artefacts.

Mean building height

Vertical morphology of the urban fabric, weighted by the surface area of each building.

Roughness class

Resistance of the urban fabric to airflow, adapted from the Davenport classes (Grimmond & Oke, 1999).

03

Results

Thermal distribution, LCZ classification, and urban heat island intensity by typology.

RESULTS

Observed land surface temperature

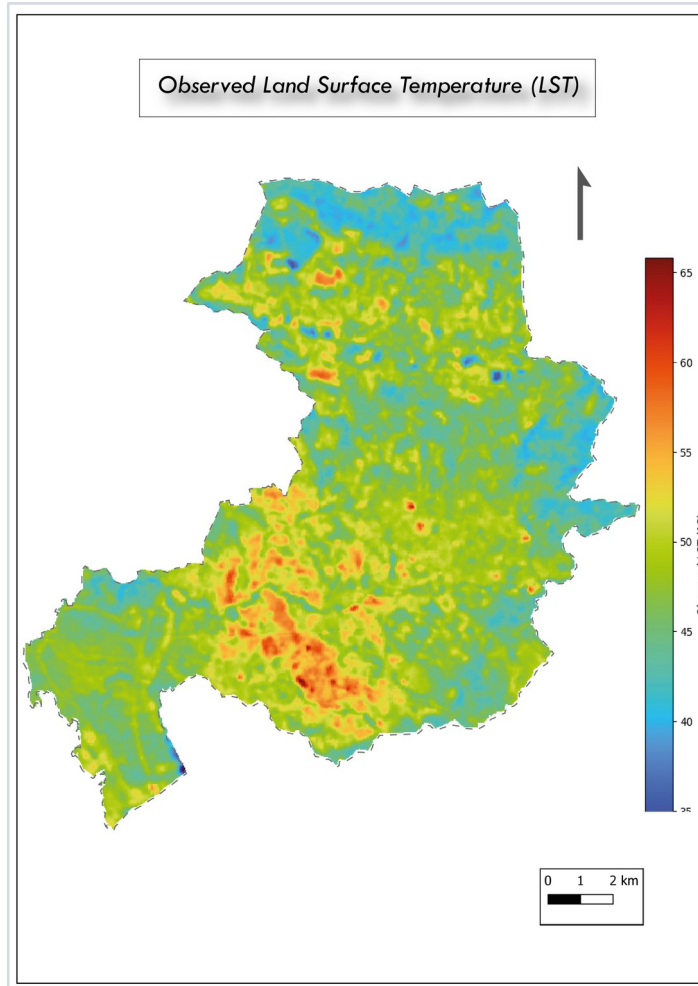


Figure 3 — Map of observed land surface temperature (LST), 30 m grid

LST values extracted over the commune show a unimodal distribution centred between 47 and 48 °C, with a range from 39 to 65 °C — consistent with the heatwave conditions confirmed on 16 August 2025.

The warmest surfaces (red/orange) concentrate in the dense urban core and in large low-rise commercial/industrial zones to the south, while cooler surfaces (blue/green) correspond to forested and agricultural areas in the north.

Source: Landsat 8 ST_B10, Collection 2 Level-2, 16 August 2025.

RESULTS

A spatial organization consistent with the urban fabric

The LCZ classification at 30 m resolution reveals a spatial organization consistent with the commune's known urban structure: classes A/B (trees) and D (low plants) very largely dominate the territory, while dense built classes remain concentrated in the urban core.

Notable result: the LCZ 1 class (compact historic centre) only emerges at this fine resolution — 56 cells — whereas it was entirely diluted in a coarser 100 m grid tested during the exploratory phase.

42.4 %

of the commune classified as LCZ A/B (dense / scattered trees)

33.3 %

of the commune classified as LCZ D (low plants)

56 cells

LCZ 1 (compact historic centre), invisible at 100 m resolution

RESULTS

LCZ classification map

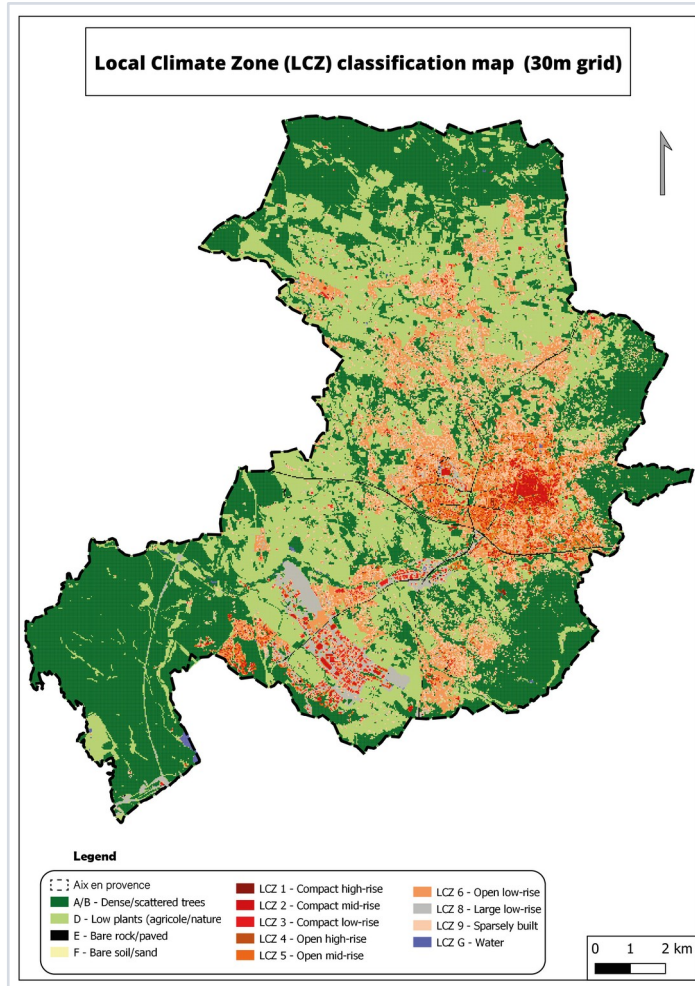


Figure 4 — Local Climate Zone (LCZ) classification map, 30 m grid

Dark green (A/B — trees) and light green (D — low plants) dominate the territory. Built classes concentrate around the historic centre (dark red cluster) and along the main road corridors, with a distinct grey cluster of LCZ 8 (large low-rise) commercial/industrial buildings to the south.

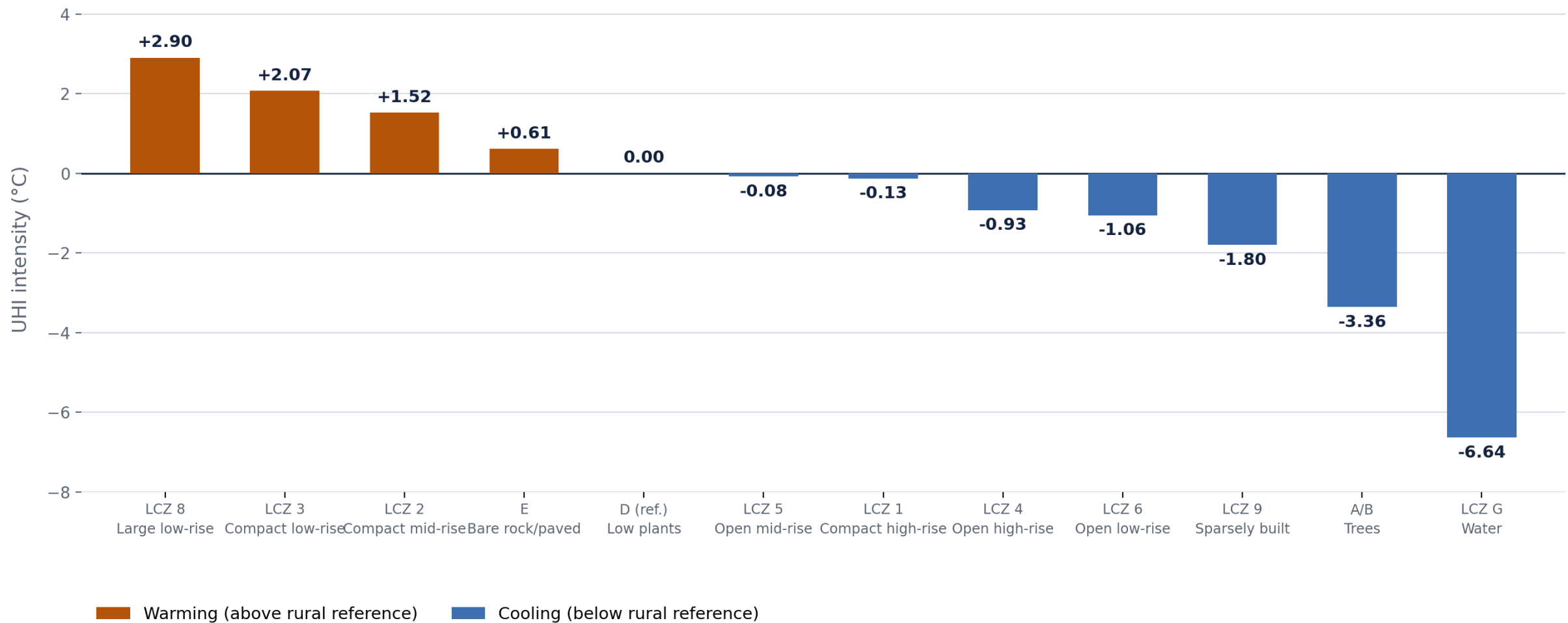
This map is the input to the cross-analysis with observed LST used to compute UHI intensity by class on the following slides.

Classification derived from BD TOPO morphological thresholds, corrected with Corine Land Cover and BD TOPO hydrography.

RESULTS

Urban Heat Island Intensity by LCZ Class

Aix-en-Provence, 16 August 2025 heatwave — reference: LCZ D (49.04 °C)



DISCUSSION

The historic centre paradox (LCZ 1)

-0.13 °C

Surface UHI intensity of the LCZ 1 class

Initial hypothesis:

the dense, tall historic centre (LCZ 1) was expected to be the hottest area of the commune.

Observed result:

a mean temperature of 48.91 °C, i.e. a near-zero UHI intensity — markedly lower than that of the LCZ 8 class (+2.90 °C).

Explanation: the divergence between surface (SUHI) and canopy-layer urban heat islands

The satellite only captures radiation emitted by surfaces visible from space (rooftops, street canyon tops), not the heat trapped within narrow urban canyons.

Shading cast by tall buildings on adjacent streets at the time of satellite overpass (10 a.m.–11 a.m.) cools the fraction of surface visible to the sensor.

The nature of roofing materials in the historic centre (tiles, stone) differs from the flat, dark roofs typical of the LCZ 8 class.

RESULTS

Explanatory factors of surface temperature

The multiple linear regression model (OLS) explains 24.7% of the spatial variance of surface temperature ($R^2 = 0.247$; $n = 210,036$; $p < 0.001$).

The moderate coefficient of determination indicates that a substantial share (75.3%) of the thermal variance remains unexplained by morphological indicators alone — consistent with the multifactorial nature of the urban heat island (albedo, thermal admittance, sky view factor not integrated).

RESULTS

Surface temperature modelled from morphology

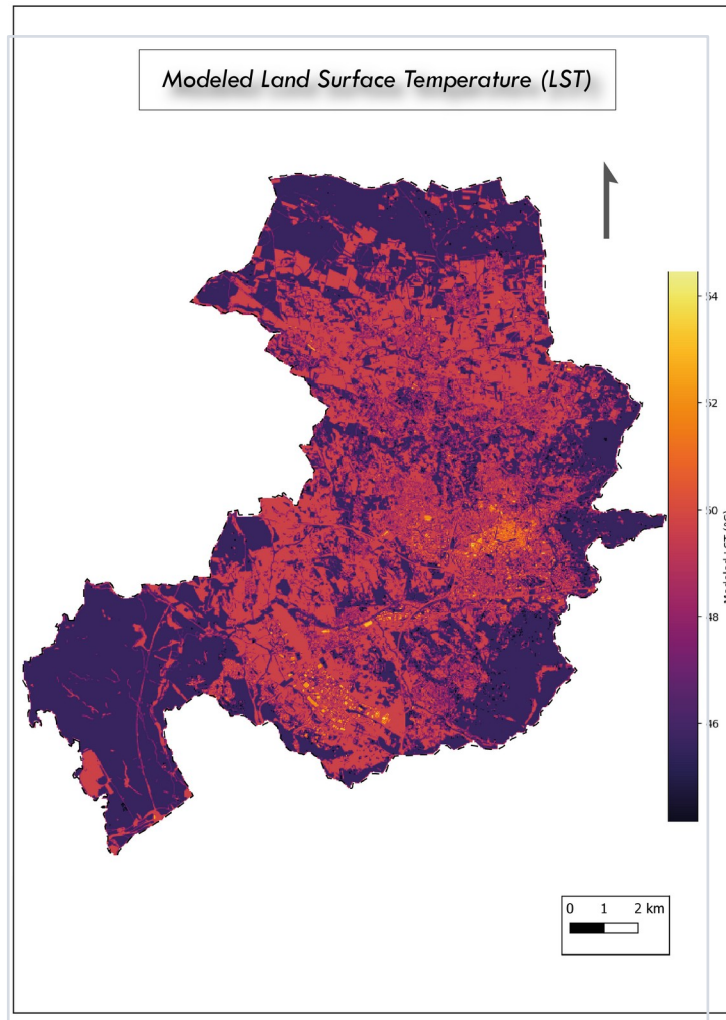


Figure 6 — Map of surface temperature modelled from morphological indicators, 30 m grid

Reconstructed solely from the five morphological variables of the regression model, this map reproduces the main spatial contrasts observed on the actual LST map, with a clear hot spot over the dense urban core and the southern industrial zone.

A smoothing effect is visible: extreme values, both high and low, are attenuated, since the model only captures the share of variance explained by the five retained indicators ($R^2 = 0.247$).

Model: OLS regression on building fraction, vegetation fraction, impervious fraction, mean height, and roughness class.

RESULTS

Regression residuals: where the model falls short

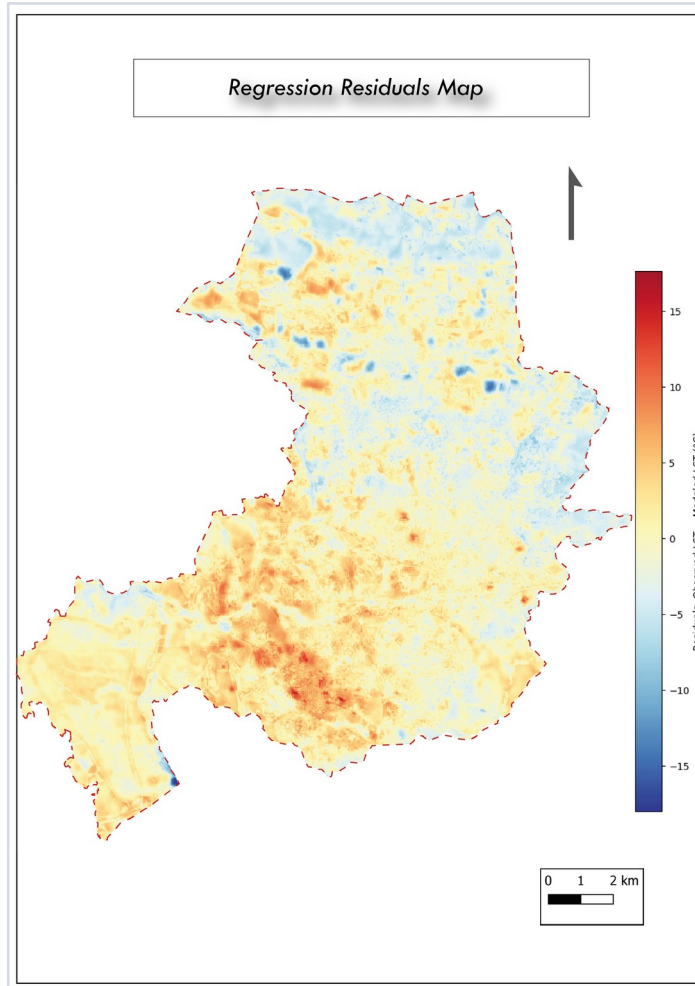


Figure 7 — Regression residuals map (observed LST – modelled LST), 30 m grid

Positive residuals (orange/red) mark cells where the model underestimates observed temperature; negative residuals (blue) mark overestimation.

The strongest positive residuals concentrate south of the urban core, over large low-rise industrial buildings and parking areas — surfaces whose flat, dark roofing and total sun exposure are not fully captured by the five morphological indicators alone.

Residuals highlight the share of thermal variability not explained by building density, vegetation, imperviousness, height, and roughness.

Methodological limitations

Sky view factor not integrated

Would require a digital surface model (DSM) including buildings and vegetation, unavailable within the reference layers used.

Surface thermal admittance

Not integrated, as it would require a correspondence table between construction materials and reference thermal admittance values.

A single observation date

16 August 2025 does not allow verification of the temporal stability of the results obtained.

No field validation

No in-situ measurement campaign (air or surface temperature) to independently validate satellite results, notably the SUHI/canopy divergence hypothesis.

CONCLUSION

Contributions and perspectives

This study mapped and characterized the urban heat island of Aix-en-Provence at 30 m resolution, revealing a thermal gradient consistent with morphological structure and the emergence of a thermally singular historic centre.

Levers for municipal urban planning

Target the LCZ 8 class (commercial/industrial zones): de-sealing, reflective or green roofing.

Protect and expand tree cover (LCZ A/B): the only morphological lever reducing temperature by more than 4 °C.

Do not mistake the historic centre's satellite thermal neutrality for actual pedestrian-level comfort.

Research perspectives

Computing a vegetation index (NDVI) from Sentinel-2 imagery.

Analysing a second observation date to test temporal robustness.

A field validation campaign (mobile transects or fixed sensors).

Integrating the sky view factor via a digital surface model.

Thank you for your attention

Questions and discussion

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